Bubble Sort

Also called: Sinking Sort or Exchange Sort

Bubble sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements if they are in wrong order.

EXAMPLE

FIRST PASS

3 1 5 4 2

1 3 5 4 2

1 3 4 5 2

1 3 4 2 $\boxed{5}$

With first pass, largest element will come to end of array

SECOND PASS With second pass, second largest element comes at second last index

1 3 4 2 5

1 3 2 $\boxed{4\ 5}$

Don't need to compare again and again as it is already sorted.

THIRD PASS

1 3 2 4 5

$\boxed{1\ 2\ 3\ 4\ 5}$

SORTED!!

COMPLEXITY

- **Space Complexity:** O(1) constant
  - ➔ Since no extra space is required here, i.e. no copying and pasting arrays etc. is not required. ➔ Also known as "inplace sorting algorithm".

- **Time Complexity:**

① **BEST CASE:** Array is sorted

$i = 0$ first pass

1 2 3 4 5 → It ran only once and we don't need to check it again.

NOTE: When j never swaps for value of i, it means array is sorted. Hence, you can end the program.

$\boxed{\text{BEST CASE COMPARISONS} = N-1 \Rightarrow N}$

CONSTANTS ARE IGNORED

② WORST CASE: Sorting descending order array to ascending order.

$i = 0 \Rightarrow$ First Pass

5, 4, 3, 2, 1 → 4, 5, 3, 2, 1 → 4, 3, 5, 2, 1 → 4, 3, 2, 5, 1 → 4, 3, 2, 1, 5 (N-1) swaps

$i = 1 \Rightarrow$ Second Pass

4, 3, 2, 1, 5 → 3, 4, 2, 1, 5 → 3, 2, 4, 1, 5 → 3, 2, 1, 4, 5 (N-2) swaps

$i = 2 \Rightarrow$ Third Pass

3, 2, 1, 4, 5 → 3, 2, 1, 4, 5 → 2, 3, 1, 4, 5 → 2, 1, 3, 4, 5 (N-3) swaps

$i = 3 \Rightarrow$ Fourth Pass

2, 1, 3, 4, 5 → $\boxed{1, 2, 3, 4, 5}$ (N-4) swaps

$$\text{total comparisons} = N-1 + N-2 + N-3 + N-4$$
$$= 4N - (1+2+3+4)$$
$$= 4N - \left[\frac{N \times (N+1)}{2}\right]$$
$$= 4N - \frac{N^2+N}{2}$$
$$= O\left[\frac{7N - N^2}{2}\right] = \boxed{O(N^2)}$$

- STABLE AND UNSTABLE SORTING ALGORITHM:

ORIGINAL ARRAY	10	20	20	30	10
			↓ AFTER SORTING		
STABLE SORTING	10	10	20	20	30
UNSTABLE SORTING	10	10	20	20	30

- In stable sorting algorithm, sorting order is maintain whereas it is the opposite in unstable sorting algorithm.